MILITARY SPECIFICATION

FLAVORED DAIRY DRINK, DRY, CHOCOLATE-COFFEE FLAVORED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a dry, fat-containing, chocolate-coffee flavored milk product which is readily dispersible in cold water, and is intended for use by the Armed Forces as a component of operational rations (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

HHH-C-575	-	Coffee, Instant
JJJ-C-501	-	Cocoa
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-M-1036	-	Milk Fat (For Recombined Milk and Other Manufactured Dairy Products)
MIL-M-3233	-	Milk Fat, Anhydrous
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-M-35052	-	Milk, Nonfat, Dry
MIL-L-35078	-	Loads, Unit; Preparation of Nonperishable Subsistence

FSC 8910

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STANDARDS

FEDERAL

FED-STD-101 - Preservation, Packaging and Packing Materials: Test Procedures
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-671 - Sanitary Standards for Milk Evaporating and Drying Plants

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers

Bacteriological Analytical Manual

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

U.S. Department of Agriculture

Directory of Consumer Protection Programs, Establishments, Circuits and Officials

(Application for copies should be addressed to the U. S. Department of Agriculture, Information Division, Consumer and Marketing Service, Washington, D. C. 20250).

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USDA Standards for Grades of Dry Whole Milk

General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service

Dairy Plants Surveyed and Approved for USDA Grading Service

(Application for copies should be addressed to the U. S. Department of Agriculture, Dairy Division, Consumer and Marketing Service, Washington, D. C. 20250.)

Brucellosis Eradication - Recommended Uniform Methods and Rules, Agricultural Research Service, U.S. Department of Agriculture, Bulletin ARS 91-79.

(Application for copies should be addressed to the U.S. Department of Agriculture, Agricultural Research Service, Washington, D. C. 20250).

Technical Association of the Pulp and Paper Industry

Testing Methods, Recommended Practices and Specifications

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 360 Lexington Ave., New York, N. Y. 10017.)

American Public Health Association

Standard Methods for the Examination of Dairy Products

(Application for copies should be addressed to the American Public Health Association, 1790 Broadway, New York, N. Y. 10019.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D. C. 20044.)

American Dry Milk Institute, Inc.

Standards for Grades of Dry Milk Including Methods of Analysis, Bulletin 916

(Application for copies should be addressed to the American Dry Milk Industry, Inc., 130 No. Franklin St., Chicago, Ill. 60606.)

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Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Boulevard, Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

3. REQUIREMENTS

* 3.1 Material.- The material shall comply with table I.

TABLE I.- Material

Item	Requirement
1	The raw milk used in the preparation of this product shall comply with all the quality requirements for No. 2 (or better) specified in the General Specification for Dairy Plants Approved for UDSA Inspection and Grading Service. Undergrade milk shall not be used. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U. S. Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in: (1) A Modified Accredited Tuberculosis Area; and (2) Either (a) A certified Brucellosis-Free Area; or (b) Modified Certified Brucellosis Area; or (3) An area in the process of being accredited or certified by the USDA in accordance with USDA Agricultural Research Service Bulletin 91-79. The milk shall be practically free from colostrum, fresh, wholesome, and normal in appearance and odor and shall be subject to inspection by the procuring agency or duly authorized representative.

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TABLE I.- Material (cont'd)

Item	Requirement
2	Dry whole milk shall comply with the USDA Standards for extra on premium grade. A USDA grade certificate shall be furnished for the grade of product used. The dry whole milk shall be not more than 30 days old from date of pack to time of use in the product. In addition, the dry whole milk shall be negative when tested for Salmonella.
3	Nonfat dry milk shall conform to the commodity requirements for type I, style B, or type II, style A product of MIL-M-35052. At time of use, in the product, the nonfat dry milk shall be not more than 60 days old following the initial drying of the fluid milk.
4	Skim milk, concentrated skim milk, and cream shall be obtained from whole milk complying with the requirements of item 1 of this table.
5	Milk fat from plastic cream or churned fat shall conform to the commodity requirements of MIL-M-1036.
6	Anhydrous milk fat shall conform to the commodity requirements of MIL-M-3233.
7	Sugar shall be white, refined cane or beet sugar, or a combination thereof, either dry or in liquid form.
8	Cocoa powder shall comply with the requirements for type II, style B of JJJ-C-501.
9	Instant coffee shall comply with the requirements for type I of HHH-C-575.
10	Vanilla flavoring shall consist of ground vanilla beans (vanilla powder), vanilla extract, imitation vanilla extract, synthetic vanillin or ethyl vanillin or combinations thereof. When the flavoring is in the dry form, appropriate carriers such as lactose, dextrose, sucrose, starch, or tricalcium phosphate may be used.
11	Stabilizers, emulsifiers or antioxidants, when used, shall be in accordance with 3.7 and shall be limited to the quantity which will accomplish their intended purpose.

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* 3.2 Processing.- The product shall be processed in accordance with either Method I or Method II of table II.

TABLE II.- Processing

Item	Requirement
	Method I
1	Incoming milk, unless processed within 2 hours following receipt, shall be cooled and held at 45°F. or lower until processing starts. Milk which must be held longer than 1 hour between stages of processing shall be cooled to 45°F. or lower and held at this temperature until processing resumes. Bulk milk in storage tanks within the dairy plant shall be handled in such a manner as to minimize bacterial increase.
2	A liquid slurry shall be prepared by using a combination of ingredients specified in table I, to produce a finished product meeting the requirements of 3.3. The liquid slurry shall be pasteurized prior to condensing (if required) and drying. Pasteurization shall be accomplished by one of the following methods: (a) heating to not less than 155°F. and holding at this temperature for not less than 30 minutes, (b) heating to not less than 171°F. and holding at this temperature for not less than 15 seconds, or (c) any other pasteurization process recognized as equally efficient by the U. S. Public Health Service and approved by the State health authority.
3	The slurry shall be dehydrated by spray drying or by any other process giving a dry product equal to that obtained by spray drying. Spray driers shall be of the continuous product removal type. Wet collectors shall not be used. Upon removal from the drier, the product shall be cooled to 110°F. or lower prior to bulk storage or final packaging. The product shall be instantized either by a single step spray-drying process or by partially rehydrating and redrying the product.
4	The sucrose vanilla and coffee may be dry blended into the product, if addition to the liquid slurry will create a problem during spray drying.

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TABLE II.- Processing (cont'd)

Item	Requirement
	Method II
5	A dry blend formulation consisting of the following composition:

Ingredients	Percent
Dry whole milk	84.60
Nonfat dry milk	2.50
Sucrose	8.50
Cocoa	2.30
Powdered vanilla	1.50
Instant coffee	0.25
Stabilizers, emulsifiers, or antioxidants, as needed	0.35

* 3.3 Finished product.- The finished product shall comply with table III.

TABLE III.- Finished product

Item	Requirement
	Physical requirements
1	The color shall be light chocolate.
2	The appearance shall be uniform in composition, and shall be free from lumps that do not fall apart under light pressure.
3	The product shall be free from foreign material such as glass, paint, wood, metal, rust, hair, insects, insect parts or other extraneous material.
4	The reconstituted product (see 4.5.5) shall have a palatable chocolate-coffee flavor and shall be free from objectionable flavors and odors, such as scorched, harsh, bitter or rancid.

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TABLE III.- Finished product (cont'd)

Item	Requirement	
	Analytical requirements	
5	Moisture, not more than	2.4 percent
6	Fat (milk fat and cocoa fat), not less than	23.0 percent
7	Protein, not less than	23.0 percent
8	Sucrose, not less than	8.5 percent
9	Oxygen (headspace), not more than	2.0 percent
10	Iron 1/, not more than	15.0 parts per million (ppm)
11	Copper 1/, not more than	1.5 ppm
12	Dispersibility, not less than	49.0 grams (g.)
13	Phosphatase, less than 2/	1 microgram of phenol per ml. of product
	Microbiological requirements	
14	Standard plate count, not to exceed	20,000 per g.
15	Coliform count, not to exceed	10 per g.
16	Salmonella	Negative

1/ If copper and iron tests made on the initial production lot comply with the requirements, thereafter, the supplier need only provide actual tests every 6 months to satisfy the copper and iron requirements. If equipment changes have been made since the initial production lot (including sanitary and recirculating piping), testing for copper and iron shall be resumed and performed once. In any case of noncompliance with requirements, testing of subsequent lots shall be continued until 2 successive lots are in compliance.

2/ Reconstitute in the ratio of 4 ounces of product and 16 ounces of water.

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3.4 Age of product.- The product shall be shipped within 60 days following manufacture.

* 3.5 Prepackaging holding requirements.- The finished product may be held for not more than 7 days after manufacture before final packaging. If not packaged in the final container within 24 hours after manufacture, the product shall be continually stored either in an atmosphere of inert gas or under a constant vacuum of not less than 20 inches. If the storage container is opened or the vacuum broken more than 24 hours prior to the final packaging, either of the above conditions shall be maintained. Dry blending of the product more than 24 hours prior to final packaging (see 3.2) shall not be permitted unless the above operations are repeated. The storage containers shall not have surfaces of iron or copper exposed to the product.

3.6 Plant qualification.- The product shall be prepared, processed and packaged in plants which operate under USDA Regulations Governing Grading and Inspection of Manufactured or Processed Dairy Products and listed by the U.S. Department of Agriculture in the document titled "Dairy Plants Surveyed and Approved for USDA Grading Service" and approved for the finished product described in this specification or in plants which have a pasteurization plant compliance rating of 90 or more as certified by a State Milk Sanitation rating officer and listed in the document titled "Sanitation Compliance and Enforcement Rating of Interstate Milk Shippers" or in plants meeting the requirements of MIL-STD-671.

3.7 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Plant qualification conditions.- The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.6.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.1.1 Inspection of milk and milk products.-

4.3.1.1.1 Animal sources.- Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in disease-free herds as specified in table I, item 1. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest office of the Agricultural Research Service of the Animal Health Division. (Location of this office may be ascertained by consulting the publication entitled Directory of Consumer Protection Programs, Establishments, Circuits and Officials.) Nonconformance to the above referenced requirements shall be cause for rejection of the lot.

4.3.1.1.2 Establishment sources for milk and milk product components.- Inspection shall be made to determine that all processed dairy products used in the manufacture of the end item originate from Military approved sources, as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement. Alternatively, approved establishment sources shall be plants having a pasteurization plant compliance rating of 90 or more as certified by a state milk santitation rating officer and listed in the document entitled: Sanitation Compliance and Enforcement Ratings of Interstate Milk Shippers published quarterly by the U. S. Public Health Service, or establishments listed by the U. S. Department of Agriculture in the document entitled: Dairy Plants Surveyed and Approved for USDA Grading Service and approved for the specific product used in the manufacture of the end item. Nonconformance to the above requirement shall be cause for rejection of the involved component lot or finshed product made therefrom, as applicable.

* 4.3.1.2 Examination of raw whole milk.- Examination and testing of raw whole milk shall be made to determine compliance with the requirements of item 1, table I, as concerns appearance, odor and referenced bacterial and sediment standards at the approximate time of use. Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more of the above referenced requirements, reflected by actual examination and tests or by examination of records shall be cause for rejection of the involved lot of finished product.

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* 4.3.1.3 Examination of dry whole milk and nonfat dry milk.- Compliance of dry whole milk with the requirements of item 2, table I, as concerns USDA grade, age and salmonella status shall be ascertained by examination for the presence of USDA grade certificate and certificate attesting to the product being salmonellae negative and marked date (or code) of pack. Compliance of nonfat dry milk with the requirements of item 3, table I shall be determined in accordance with the applicable examination criteria for product characteristics contained in the referenced specification except that the sampling plans and acceptable quality levels (AQLs) shall not apply. The sample unit for examination shall be approximately 8 ounces of product derived from each of 5 primary containers or all containers if less than 5 form a lot. Failure to comply with one or more of the above referenced requirements or missing information with regard to age of product and applicable certificates shall be cause for rejection of the involved quantity of finished product made therefrom.

* 4.3.1.4 Examination of skim milk, concentrated skim milk and cream.- Examination of skim milk, concentrated skim milk and cream shall be made to determine compliance with the requirements of item 4, table I. This examination shall be performed by a review of plant records, laboratory reports or other valid documents, as applicable. In addition, examination shall made of these components to determine compliance with the condition requirements of normal appearance and odor reflected by the original milk. The sample unit for examination shall be approximately 6 ounces derived from each of 5 primary containers or all containers if less than 5 form a lot. Failure to comply with the above referenced requirements, reflected by actual examination or by examination of records, shall be cause for rejection of the involved lot of finished product made therefrom.

* 4.3.1.5 Examination of milk fat (from plastic cream or churned fat), anhydrous milk fat, cocoa powder and instant coffee.- These ingredients shall be examined in accordance with the applicable examination criteria for product characteristics contained in the specifications referenced in items 5, 6, 8 and 9 of table I, except that the sampling plans and AQLS shall not apply. The sample for examination shall be approximately 6 ounces of product derived from each of 5 primary containers or all containers if less than 5 form a lot. Use of these ingredients having one or more defects shall be cause for rejection of the involved quantity of finished product made therefrom.

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* 4.3.1.6 Examination of sugar, vanilla flavoring, stabilizers, emulsifier and antioxidants.- Examination of these components to determine compliance with the identity requirements of items 7, 10 and 11 of table I shall be ascertained by a review of appropriate invoices, labels or other valid documents, as applicable. Noncompliance with one or more identity requirements, referenced above, shall be cause for rejection of the involved quantity of finished product made from such components.

* 4.3.2 Testing of components.- Testing of components described in this specification shall be as shown in table IV. Results shall be applicable to the lot average. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE IV.- Testing of components

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to	Test ref and reqmt para
				Thickness of:		
Flexible bag	1 running yard or equivalent in bags	Rolls or bags	S-1	Polyolefin or food grade vinyl	Nearest 0.001 inch 1/	5.1
				Aluminum foil	Nearest 0.0001 inch 1/	
				Polyester	Nearest 0.0001 inch 1/	
Paperboard	Two 12 x 12 inch panels	Boxes	S-1	Thickness of paperboard	Nearest 0.001 inch	5.1.1
				Bursting strength	Nearest pound per sq. inch	

1/ In lieu of performing test for thickness of each lamina, conformance to thickness requirements may be ascertained by examination of pertinent labels, invoices or other valid documents. If nonconformance is indicated, the involved quantity of finished product shall be rejected.

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* 4.3.3 Process examination.-

4.3.3.1 Age of components (as applicable), temperatures, time and time intervals, equipment, formulation and procedures.- Examination shall be performed during processing to determine conformance to the requirements of items 2 and 3 of table I pertaining to age of product at time of use and of table II as concerns pertinent processing criteria. Records of time, temperature, formulation and all applicable control data shall be maintained. Nonconformance to one or more of the above referenced requirements, reflected by actual examination or by records, shall be cause for rejection of the involved quantity of finished product.

4.3.3.2 Examination prior to packaging (when applicable).- The finished product shall be examined prior to final packaging (when held for more than 24 hours after drying, see 3.5) to determine compliance with requirements as concerns time of storage, inert gas atmosphere, vacuum, dry blending and type of storage container. Storage time, time intervals, vacuum and time of blending records shall be maintained and made available for review. Nonconformance to one or more of the above referenced requirements, as reflected by actual examination or by records, shall be cause for rejection of the lot.

* 4.3.4 Examination of end item.- Classification of defects found during examination shall be in accordance with tables VI through X and 4.3.4.1 through 4.3.4.4. The examination criteria for the above tables shall be as shown in table V. The AQL's shall be expressed in terms of defects per hundred units for tables VI, VIII, IX and X, and as percent defective for table VII.

TABLE V.- End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs			
				Major A	Major B	Minor	Total
VI	S-4	1 bag	Bags	0.40	1.5	—	4.0
VII	S-3	1 bag	Bags	—	—	2.5	—
VIII	S-3	1 bag	Bags	—	0.65	—	—
IX	S-2	1 bag	Bags	1/	1.5	—	6.5
X	S-3	1 intermediate carton	Intermediate cartons	—	—	4.0	—

1/ See footnote 1/ to table IX.

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TABLE VI.- Examination of flexible packages (see 5.1)

Category			Defect
Major A	Major B	Minor	
101			Open seam or seal, tear, hole or sifter.
	151		Not type, style or size specified.
	152		Not heat sealed as specified.
	153		Evidence of delamination.
	154		Exterior color of bag not as specified. 1/
	155		Unclean.
	156		Nomenclature or directions for use missing, incorrect or illegible.
		201	"V" shaped notch missing in one or both side seals, or not located as specified.
		202	"V" shaped notch less than 1/16 or more than 1/8 inch deep.
		203	Required label information other than nomenclature and directions for use, missing, incorrect or illegible.
		204	Contrasting color for labeling wording not used or location of labeling not as specified.

1/ White mark for electric-eye positioning, when used, shall be acceptable.

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TABLE VII.- Examination for net weight 1/ 2/

Category	Defect
Minor	
201	Net weight less than 3-3/4 ounces.

1/ The lot shall be rejected if sample data indicate lot average net weight is less than specified net weight.

2/ Report to the nearest 1/8 ounce.

TABLE VIII.- Examination for leakage 1/

Category	Defect
Major B	
151	Steady progression of bubbles.
152	Evidence of delamination. 2/

1/ See 4.5.1.1 for procedure.

2/ Examine after removal from water.

TABLE IX.- Examination for product characteristics 1/

Category			Defect
Major A	Major B	Minor	Dry product
101			Presence of foreign material, e.g. glass, paint, wood, metal, rust, hair, insect, insect parts, etc.
	151		Appearance not uniform or presence of lumps that do not fall apart under light finger pressure.
		201	Not a light chocolate color.

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TABLE IX.- Examination for product characteristics 1/ (cont'd)

Category			Defect
Major A	Major B	Minor	Reconstituted product 2/
102			Presence of objectionable flavors and odors, e.g. scorched, harsh, bitter, rancid, etc.
		152	Does not have a palatable chocolate-coffee flavor.

1/ Presence of any Major A defect shall be cause for rejection of the involved lot of finished product.

2/ See 4.5.5 for procedure.

TABLE X.- Examination of intermediate carton

Category	Defect
Minor	
201	Carton and dimensions not as specified.
202	Count less than specified.
203	Arrangement not as specified.
204	Top inner flaps not at random, and bottom inner flaps not notched for interlocking. 1/
205	Tucks less than 1 inch in length and not notched for locking. 1/

1/ Not applicable when the alternate method for intermediate cartons is utilized, in which case the applicable examination criteria in PPP-B-566 shall apply.

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* 4.3.4.1 Examination of case liner for levels A and B packing.- Examination of the case liner shall be in accordance with the examination criteria contained in the appendix of MIL-L-10547.

* 4.3.4.2 Examination of shipping container.- Examination of the filled, closed and reinforced shipping container shall be made in accordance with the applicable examination criteria in the appendix to PPP-B-636. In addition, the following defects shall be included in the examination: Major: (1) markings missing, incorrect or illegible; (2) count of intermediate cartons less than specified; Minor: (1) arrangement not as specified; (2) chipboard pad missing (when applicable). For level B pack, when applicable, reinforcement of the shipping container with other than pressure-sensitive adhesive filament-reinforced tape shall not be acceptable. When packing for level C is required to be in accordance with 5.2.3.1, examination shall be in accordance with the aforementioned specification, except that only the defects for marking and arrangement shall apply.

* 4.3.4.3 Examination of unit loads.- Unit loads shall be examined in accordance with the examination criteria of MIL-L-35078.

* 4.3.4.4 Examination of age of product.- Examination shall be made at time of delivery to determine conformance to the requirements of 3.4, as concerns age of product. Determination of compliance shall be ascertained by marked date of pack, codes or other valid means. Nonconformance to the above referenced requirements shall be cause for rejection of the lot.

* 4.4 Sampling procedures and acceptance criteria for testing of the finished product.- The finished product shall be tested for the analytical and microbiological requirements specified in 3.3, table III. Procedures for testing shall be in accordance with 4.5. (For exceptions to iron and copper testing, see footnote 1/ to table III.) Test requirements for bacteria, coliform, salmonella and oxygen in the headspace shall be on a unit basis. All other tests shall be on a lot average basis. The sample unit for bacteria, coliform, oxygen and salmonella shall be one primary container. Lot size shall be expressed in terms of primary containers. Sample size shall be the number of primary containers indicated by inspection level S-1. The sample for testing for total fat, moisture, protein, sucrose, iron, copper, dispersibility and phosphatase shall be approximately a 1-pound composite derived from the containers used for unit testing. If the sample size is insufficient to yield the desired amount of composite sample, additional primary containers shall be taken at random. The results shall be reported as follows: Moisture, fat, protein, sucrose and oxygen to the nearest 0.1 percent; iron and copper to the nearest 0.1 ppm; dispersibility to the nearest 0.1 gram; standard plate and coliform counts in accordance with Standard Methods for the Examination of Dairy Products; salmonellae as negative or positive, and phosphatase as pass or fail. Nonconformance to one or more defects of the above referenced requirements shall be cause for rejection of the involved lot of finished product.

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* 4.5 Test methods.- Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.5.1 Examination procedures.-

4.5.1.1 Leakage test.- The flexible package shall be tested for leakage by submerging the bag in water contained in a vacuum desiccator or similar device and drawing a vacuum of 10 inches of mercury for 30 seconds. A steady progression of bubbles from any portion of the bag will constitute a leak. Isolated bubbles caused by entrapped air will not be considered as signs of leakage.

4.5.2 Test procedures.-

4.5.2.1 Chemical analyses.- Chemical analyses shall be made in accordance with the methods cited in 4.5.2.2 and 4.5.2.3.

4.5.2.2 The following tests shall be conducted as specified in Official Methods of Analysis of the Association of Official Analytical Chemists:

Item	Test	Source	Method
1	Titratable acidity	1/	Acidity weigh sample)
2	Total fat	2/	Fat in dried milk
3	Protein	2/	% Protein = (% N X 6.38)
4	Sucrose	3/	Sucrose
5	pH of cocoa, when required	4/	Hydrogen-Ion Concentration; Electrometric Method
6	Cacao fat, when required	5/	Fat, Quantitative Determination, Method II

1/ Chapter: Dairy Products; Section: Milk

2/ Chapter: Dairy Products; Section: Dried Milk, Nonfat Dry Milk, and Malted Milk

3/ Chapter: Dairy Products; Section: Sweetened Condensed Milk

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4/ Chapter: Cereal Foods; Section: Wheat Flour

5/ Chapter: Cacao Bean and Its Products; Section: Fat

4.5.2.3 The following tests shall be conducted as specified in Standards for Grades of Dry Milk Including Methods of Analysis, Bulletin 916.

Test	Source and method
Moisture	Determination of Moisture (Toluene Distillation)
Copper	Determination of Copper
Iron	Determination of Iron
Dispersibility	Determination of Dispersibility of Instant Nonfat Dry Milk. (Use 60°F. water and a U. S. No. 40 screen).

4.5.2.3.1 Oxygen content of the filled and sealed bag shall be determined in accordance with Method 6011 of FED-STD-101.

4.5.3 Microbiological and sediment analyses.- Microbiological and sediment analyses shall be made in accordance with the following methods from American Public Health Association, Inc., publication entitled "Standard Methods for the Examination of Dairy Products" (except as noted):

Test	Source and method
Standard plate count	Agar Plate Method (32°C.)
Coliform count	Coliform Bacteria (32°C.) (using Desoxycholate Lactose Agar)

4.5.3.1 Salmonella.- Salmonella test shall be determined by the procedure outlined in the Bacteriological Analytical Manual of the U. S. Department of Health, Education and Welfare, Federal Food and Drug Administration.

4.5.4 Phosphatase.- Phosphatase shall be determined in accordance with the following methods from Standard Methods for the Examination of Dairy Products, Chapter: Phosphatase Methods to Determine Pasteurization, Section: Scharer Rapid Phosphatase Test.

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4 5.5 Reconstitution.- Reconstitution shall be in the ratio of 4 ounces of dry product to 8 fluid ounces of water at a temperature of 60 ± 5°F.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging, level A, flexible package.- Four ounces of the product shall be filled into a flat-style bag with inside dimensions of 4-1/2 by 8 inches, the first dimension being the opening of the bag. The bag shall be fabricated from a laminated material, consisting of 0.003-inch thick polyolefin or food grade vinyl, laminated to 0.0005-inch thick aluminum foil, laminated on the plain side to 0.0005-inch thick polyester. The adhesive used for laminating shall be a suitable odorless and permanently plastic water-resistant compound other than wax. The complete exterior of the bag shall be colored with a color in the number range from 30045 to 30118, or from 34052 to 34108 (both inclusive) of FED-STD-595, except that a white mark may be used for electric-eye positioning. The bag shall be formed with heat-sealed seams on 3 sides. A V-shaped tear notch, 1/16 to 1/8 inch in depth, shall be made in both side seams, 1 inch ± 1/8 inch from the open end of the bag. Filling of the bag shall be under conditions which will result in an oxygen content of not more than 2.0 percent, when tested in accordance with 4.5.2. The bag shall be closed with a heat seal extending not more than 3/4 inch from the opening of the bag. The filled and sealed bag shall not leak when tested in accordance with 4.5.1.1.

5.1.1 Intermediate carton.- Ten flexible packages shall be placed in a carton in a horizontal position, 1 on top of the other. The carton shall be constructed in accordance with style II, type D, class e of PPP-B-566. The dimensions of the carton shall not exceed 8-1/2 inches in length by 5 inches in width, by 6 inches in depth. The tucks shall be at least 1 inch in length and shall be notched for locking. The top inner flaps shall be at random, and bottom inner flaps shall be notched for interlocking. Alternatively, the box may be style XI, automatic fold, as specified in PPP-B-566. Paperboard for the box shall be 0.032-inch thick and shall have a minimum bursting strength of 112 pounds per square inch when tested in accordance with Test Method T 410M of the Technical Association of the Pulp and Paper Industry.

* 5.2 Packing.- Sixteen intermediate cartons, arranged 4 in length, 2 in width, and 2 in depth, shall be packed into snug-fitting shipping containers in accordance with level A, B or C, as specified (see 6.2).

5.2.1 Level A.- The shipping container shall be a fiberboard box with liner, constructed, closed, and reinforced in accordance with style RSC-L, V2s of PPP-B-636. The intermediate cartons shall be inclosed in a waterproof case liner,

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fabricated and sealed in accordance with MIL-L-10547, for subsistence items. If flaps of the shipping container are closed with metal stitches the case liner shall be protected from the stitches by placing a chipboard pad between the case liner and the stitches. When unit loading is specified reinforcement is not required.

5.2.2 Level B.- The shipping container shall be a fiberboard box constructed, closed, and reinforced in accordance with style RSC-L, V3c, V3s, or V4s, of PPP-B-636. The intermediate carton shall be inclosed in a waterproof case liner, fabricated and sealed in accordance with MIL-L-10547 for subsistence items. The case liner shall be protected from stitches as specified in 5.2.1. Reinforcement is not required when unit loading is specified.

5.2.2.1 When specified (see 6.2 and 6.4), the shipping container specified in 5.2.2 shall be reinforced with pressure-sensitive adhesive, filament-reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C.- The shipping container shall be a fiberboard box, constructed and closed in accordance with style RSC, type CF, or SF, class domestic, method II closure, of PPP-B-636.

5.2.3.1 When specified (see 6.2 and 6.5) the shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Unit loads.- The product packed as specified in 5.2 shall be arranged in unit loads in accordance with MIL-C-35078 for the type and class of load specified (see 6.2).

5.4 Labeling and marking.-

5.4.1 Flexible packages.- The following information shall be printed on one side of the package in clearly legible type and in a contrasting color:

FLAVORED DAIRY DRINK, DRY

CHOCOLATE-COFFEE FLAVORED

Net weight
Date of pack (day, month, year)
Name and address of manufacturer

DIRECTIONS FOR USE

Aid station, hospital or dieting:

Mix contents with 1/3 canteen cup
(8 fluid ounces) hot or cold water.

Regular beverage:

Mix contents with 2/3 canteen cup
(16 fluid ounces) hot or cold water.

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5.4.2 Shipping containers.- Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Marking of unit loads.- Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Intended use and product information.- The product covered by this specification is intended primarily for use as a readily dispersible mild-flavored milk drink, when reconstituted with the prescribed amount of cold or hot water to an approximate total solids level of 32 percent, for use in aid stations as a high-protein, high-caloric drink for casualties, or as a general feeding supplement in hospital feeding. It may also be used for general troop feeding as a regular flavored milk drink, by reconstituting the product to a 16 to 18 percent total solids level. Prior to reconstitution refrigeration is not necessary; however, stability is improved if the product is stored in a cool location. After reconstitution the product should be handled as fresh milk.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing required (see 5.2).
(c) When packing specified in 5.2.1or 5.2.2.1 is required.

6.3 Packing specified in 5.2.1 is intended to provide the maximum military protection required for open storage of unit loads in areas of heavy rainfall, high humidity, and adverse weather conditions.

6.4 Packing specified in 5.2.2.1 is intended for transfer at sea operations and specific overseas operations.

6.5 Award of contract for the product specified in this document shall be limited to plants known to maintain the required sanitation conditions of 3.6 (also see 4.2).

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP- SS

Preparing activity:

Army - GL

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